

Foreword and introduction

SGA 3

English draft

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About this draft

This is an unofficial English translation of the foreword and the introduction from *Schémas en groupes* (SGA 3). The source is the recomposition edited by P. Gille and P. Polo, PDF `AaIntro-18fev11.pdf` (6 pages), from <https://webusers.imj-prg.fr/~patrick.polo/SGA3/>. The seminar was directed by M. Demazure and A. Grothendieck (IHÉS, 1962–1964).

Statement numbers follow that PDF. A clear typographical slip is corrected in the English and marked with a `% typo:` comment; such slips are listed in the README. A slip that would change the mathematics is left as printed and marked `% typo?:`. Scholarly proofreading remains outstanding.

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Contents

About this draft	iii
Foreword and introduction	1
Bibliography	5

Foreword and introduction

Foreword

We present here a slightly revised re-edition of the original Seminar, whose aim and content are indicated in the Introduction. The revision consisted essentially in the correction of misprints, the addition (in footnotes) of a few supplementary remarks or references, the present division into three volumes each provided with a detailed table of contents and an index of notations, and the addition of a terminological index at the end of volume 3. Moreover, exposé VI_B of J.-E. Bertin has been partially rewritten by him, notably paragraphs 5 and 10, so that certain references to this exposé differ from the references to the original exposé. The reader will find a list of the exposés of the Seminar at the beginning of the present volume.

Since the publication of the first edition of the present Seminar, the whole of the *Éléments de Géométrie Algébrique*, Chap. IV, has appeared, which renders certain passages of the Seminar unnecessary; we have sometimes indicated in footnotes the relevant references to EGA IV which make it possible to bypass such passages.

For another exposé on algebraic groups making systematic use of the language of schemes, we mention the book of M. Demazure and P. Gabriel, *Groupes algébriques* (North-Holland & Masson et Cie). Unlike the present Seminar, this book assumes no knowledge of algebraic geometry, but contains all the necessary preliminaries of scheme theory, and its reading can therefore serve as an introduction to the study of our Seminar. (It contains, moreover, themes not covered in the Seminar, such as the structure theory à la Dieudonné of commutative affine algebraic groups, in *loc. cit.* Chap. V.)

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Introduction

1. The aim of the present seminar is twofold.

On the one hand, we aim to give convenient foundations for the theory of group schemes in general. Exposés I to IV will give, in this respect, the indispensable preliminary exercises in schematic and categorical syntax. In order to obtain a language which “sticks” without effort to the geometric intuition, and to avoid circumlocutions which are unbearable in the long run, we always identify a prescheme X over another S with the functor $(\text{Sch})^\circ/S \rightarrow (\text{Ens})$ which it represents^(*), and it is necessary to give numerous definitions in such a way that they apply to arbitrary functors, representable or not. Moreover, almost all the functors which we shall have to use will be “sheaves” (for the “faithfully flat quasi-compact topology”); exposé IV, which treats groups only incidentally, gives a sketch of the language of “localization” and of sheaves, which likewise proves very convenient in questions of representability of functors. This exposé will above all provide us, for questions of passage to the quotient, with the most convenient framework for the sequel.

Exposé V gives some general results on the existence of quotients, taken up again in exposé VI_A in the case of the quotient of an algebraic group over a field (or, more generally,

^(*)Such a point of view seems to have been envisaged for the first time eight or nine years ago, in connection with the theory of formal groups, by P. Cartier, who unfortunately did not take the trouble to make it precise and to systematize it as it deserved.

over an artinian ring) by a subgroup^(**). This latter exposé and exposé VI_B, which follows it, also contain various elementary results special to algebraic groups over a field, commonly used in the sequel.

Exposé VII studies certain facts connected with the characteristic of the base field and develops in particular, with the appropriate generality, the correspondence between radicial group schemes of height 1 and restricted Lie p -algebras.

Finally, exposé XVIII contains the generalization, in the theory of schemes, of Weil's theorem on the "birational" definition of algebraic groups.

On the other hand, we propose to generalize, to groups over an arbitrary base prescheme, the Borel–Chevalley structure theory of affine algebraic groups. It has moreover become apparent, on the occasion of the writing up of the notes of the seminar, that the affine hypothesis was unnecessary for many results of the theory. The most complete results are of course obtained in the cases of "semi-simple group schemes", or more generally "reductive" ones, with which we shall concern ourselves exclusively from exposé XIX on. Chevalley himself had already given the construction of the groups "of Tôhoku" over the ring of integers, a construction which will be taken up again in the present seminar. The principal uniqueness theorem⁽²⁾ gives a simple characterization of the "twisted" variants of these Tôhoku groups, over a base prescheme S : these are the groups which are affine and smooth over S , whose geometric fibers are connected semi-simple groups in the usual sense^(***) ⁽²⁾.

2. As in the case of the known theory over an algebraically closed field, a crucial role is played by the subtori of the group schemes under consideration. Thus the preliminary study of tori, and more generally of "group schemes of multiplicative type", (both from the intrinsic point of view and from the point of view of the subgroups of multiplicative type of a given group), takes a rather large place in this Seminar (exposés VIII to XII). Their remarkable rigidity (greater even, in certain respects, than that of abelian schemes, or of semi-simple group schemes) makes them very effective working tools for the study of certain more general groups.

3. From exposé XII on (to the exclusion of exposé XVIII, already mentioned) we shall commonly use the theory of affine algebraic groups over an algebraically closed field, which the reader will find in the Chevalley Seminar 1956, more particularly in exposés IV to IX of this Seminar. We shall also use, but to a lesser extent, the later exposés of the Chevalley Seminar, devoted to the structure of semi-simple algebraic groups. Indeed, we shall take up Chevalley's theory directly in the framework of schemes: one will see that in this way (even over a base field) the exposé gains in simplicity and in precision.

4. The principal object of the present Seminar is of course to develop techniques which apply to the study of group schemes over an arbitrary base, i.e. essentially to the study of families of algebraic groups. As such, the infinitesimal properties of such families, and in particular the case of an artinian base scheme, play an important role. These properties intervene even for the study of algebraic groups over a field k , in the case where the latter is not perfect, so as

^(**)For a more thorough study of passage to the quotient, notably by reductive groups, see the important study of D. Mumford, *Geometric Invariant Theory*, Ergebnisse der Mathematik, Bd 34, Springer 1965. Let us observe that on one important point, the terminology of this book does not agree with ours, for over a field of characteristic $p > 0$, the groups which Mumford calls "reductive"⁽¹⁾, are the smooth groups of multiplicative type in the sense of the Seminar (cf. Exp. IX). One may doubtless consider that the sense in which Mumford takes the word "reductive", which loses its meaning over a base which is not a field, was adopted by him provisionally and as a sort of makeshift (and this is also more or less what Mumford explains, for other reasons, from the second paragraph of his preface!). ⁽¹⁾ **N.D.E.** see the remarks added at the end of this Introduction.

⁽²⁾**N.D.E.** This refers to corollary XXIII.5.6, which follows easily from the uniqueness theorem for split reductive groups (cf. XXIII, th. 4.1 and cor. 5.2), given that every semi-simple S -group is a "twisted form" (for the étale topology) of a group "of Tôhoku" (cf. XXII, cor. 2.3).

^(***)This is the essential result of the thesis of M. Demazure (*Schémas en groupes réductifs*, Bull. Soc. Math. France **93** (1965), 369–413).

to be able in particular to apply the technique of descent in the non-Galois case. Among the new results obtained in this case, let us mention the existence of maximal tori and of Cartan subgroups in an arbitrary smooth algebraic group, the rationality of the variety of maximal tori, and various related results (Exp. XIV ⁽³⁾), or the correspondence between the “forms” of a semi-simple group and the principal homogeneous bundles under a suitable semi-simple algebraic group (in general not connected) (Exp. XXIV, 1.16–1.20). In general, one may say that the methods required in order to work over a base field which is not perfect are essentially those used for arbitrary base preschemes, and thereby fall outside the framework of classical algebraic geometry.

5. It did not seem useful to indicate at the head of the written exposés the date or the dates of the corresponding oral exposés of the Seminar. Let us be content to say that the order of the mimeographed exposés (from I to XXVI) corresponds well to the order of the oral exposés. Moreover, the writing of the definitive text is sometimes distinctly later than that of the oral exposé, and often differs from it rather substantially, the written text being generally more detailed and more complete (such as Exp. IV and VII_B), or even appreciably more general (such as Exp. XII or VII_B) than the oral exposé. Other written exposés correspond to no oral exposé (VI_B, VII_A, XV, XVI, XVII, and the essential part of XXVI), and were written and inserted into the mimeographed Seminar, either to provide convenient references for various other exposés (this is notably the case of VI_B), or because they constitute a natural extension of the notions and techniques already developed. One will note as a consequence that the reading of exposés VII_A, VII_B, XV, XVI, XVII is not necessary for the study of the rest of the Seminar, and in particular for the part of this Seminar devoted to reductive group schemes.

6. From the theory of schemes, we shall use above all the general language of schemes, set out in EGA I, the notions of flat morphism, étale morphism, smooth morphism, set out in SGA 1, I to V, and finally the theory of faithfully flat descent of SGA 1, VIII. We have as far as possible avoided formulating unnecessary noetherian hypotheses, which has on the other hand obliged us to replace the usual hypothesis “of finite type” by the hypothesis “of finite presentation”. For the notion of morphism of finite presentation, the reader will consult EGA IV, 1.4 and 1.6. The results of SGA 1, I to IV, stated most often in the noetherian context, will be developed in the general case in EGA IV^(****), where standard methods will likewise be developed in detail for reducing certain types of statements (involving hypotheses of finite presentation) to the noetherian case (EGA IV, paragraphs 8, 9, 11). The reader who does not wish to admit these results of EGA IV may simplify certain statements or their proof by supposing the base prescheme to be locally noetherian. He does, however, expose himself to difficulties in the cases where the proof proceeds by descent from $\hat{A}$ to A , where $\hat{A}$ is the completion of a noetherian local ring A , for this method leads one to introduce the ring (in general non-noetherian) $\hat{A} \otimes_A \hat{A}$.

7. References will be made according to the usual decimal system: the reference 5.7.11 refers to the proposition (or lemma, definition, etc.) of that name in the same exposé; in the reference XVII 7.8 the Roman numeral indicates the number of the exposé. We shall use the following sigla for our standard references:

⁽³⁾**N.D.E.** in particular, theorems 1.1 and 6.1.

^(****)Since the writing of this introduction, the four parts (§§ 1 to 21) of EGA IV have appeared.

- Bible = Séminaire Chevalley “Groupes de Lie algébriques”, 1956/58.
- EGA X , $x.y.z$ = J. Dieudonné and A. Grothendieck, *Éléments de Géométrie Algébrique*, Chap. X , statement $x.y.z$ (or subsection $x.y$, etc.).
- SGA n , X $y.z$ = Séminaire de Géométrie Algébrique du Bois-Marie, year n , statement $y.z$ of exposé X .
- TDTE = A. Grothendieck, *Techniques de descente et Théorèmes d’existence en Géométrie Algébrique*, exposés given in the Bourbaki Seminar between 1959 and 1962.

(1)

⁽¹⁾**N.D.E.** Concerning the quotient by a reductive group, the situation has evolved considerably since the writing of Note (**) by A. Grothendieck. Indeed, for an affine algebraic group over an arbitrary field k , the “good” notion, introduced by Mumford, is that of a “geometrically reductive” group. (One says, on the other hand, that G is “linearly reductive” if every rational representation of G is completely reducible, but, as pointed out in Note (**), this condition is too restrictive if $\text{car}(k) > 0$). According to the results of M. Nagata and W. J. Haboush ([Na64], [NM64], [Ha75]), the geometrically reductive k -groups are exactly the reductive k -groups in the sense of the present Seminar. For all of this, see the later editions of Mumford’s book ([MF82]). Moreover, the extension to the case of an arbitrary base of the notion of “geometric reductivity”, and of its consequences for passage to the quotient, has been carried out by C. S. Seshadri ([Se77]), and additions, due to M. Raynaud, are to be found in the article [CTS79] of J.-L. Colliot-Thélène and J.-J. Sansuc. Finally, for more recent developments concerning passage to the quotient by an algebraic group or, more generally, by a groupoid (cf. Exposé V of the present Seminar), let us mention the articles [Ko97] and [KM97] of J. Kollár and of S. Keel and S. Mori.

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